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Method paper

Method: An accurate method for detecting drinking bouts in dairy cows based on reticulorumen temperature

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ABSTRACT

This study evaluated the performances of three methods for detecting drinking bouts in dairy cows using reticulorumen temperature (RT): the 'FixT' method based on a fixed RT threshold, the 'Cow-dT' method based on a cow-day-specific RT threshold, and the 'FallST' method based on RT fall slope. We observed the drinking behaviours of 28 dairy cows equipped with reticulorumenal sensors over 96 h to create a reference dataset. A total of 730 drinking bouts were observed. We matched detected drinking bouts against observed drinking bouts to obtain the number of true-positives, false-negatives, and false-positives, and then calculated the detection performances of the three methods in terms of sensitivity (Se), positive predictive value (PPV), and F-score. The performances of the three RT-based methods (Se $\geq$ 90%, PPV $>$ 96% and F-score $\geq$ 93%) were better than those from previous work using collar-attached accelerometers, but slightly lower than methods using drinking troughs connected to electronic identification systems or methods combining accelerometers with geomagnetic sensors or with ultra-wideband location. The FallST method showed slightly better performance (highest F-score) than the FixT and Cow-dT methods. The FallST method accurately detected drinking bouts lasting more than 30 s and at least 30 min apart, with a detection time accuracy of 10 min. The models using RT curve parameters failed to predict characteristics of the drinking bouts. In conclusion, the method developed here can accurately detect drinking bouts in dairy cows using RT, but without further characterisation of the drinking bouts (e.g. duration).

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Implications

Water is essential for dairy cows, impacting both welfare and milk production. We aimed to develop an efficient method to detect drinking bouts using reticulorumen temperature sensors. Three methods were tested: two reticulorumen temperature threshold-based approaches and a novel reticulorumen temperature fall slope method. Our results showed that the reticulorumen temperature fall slope method accurately detects drinking frequency and timing but cannot determine bout duration. This method could be applied to other cattle and ruminants. This

method could be used to enhance water intake monitoring, enabling the development of anomaly alerts that could improve animal welfare and productivity.

Specifications table

Subject	Behaviour and Health Management
Type of data	Table, Figure
How data were acquired	Video recording, Reticulorumenal sensor (Large Thermobolus, ITK/NewMedria, Châteaubourg, France), Algorithm developed in Visual Basic for Applications (VBA) in Microsoft® Excel (2019),

(continued on next page)

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	The Observer XT v14 behavioural coding software (Noldus, Wageningen, The Netherlands, R 4.4 (R Core Team, 2024))
Data format	Raw, pretreated and analysed data.
Parameters for data collection	Data were collected from 28 primiparous Holstein dairy cows that were between 100 and 163 days in milk. The cows were loose-housed in a cubicle barn. The water trough was located on the feed-access and exercise area. The water temperature was around 11 °C. Cows were fed once per day with a total mixed ration. The cows were milked twice a day in a milking parlour.
Description of data collection	Data were collected by observing the drinking behaviours of 28 dairy cows equipped with reticulorumenal sensors. The cows were continuously video-recorded over two periods of 48 h. At the same time, the sensor recorded reticulorumen temperature every 5 min. The bolus data were transmitted to a local internet-connected gateway by LoRa (Long Range) technology and thence to FarmLife® web application (ITK/NewMedria, Chateaubourg, France), from where it was regularly downloaded for analysis and curation.
Data source location	Institution: Herbipôle, INRAE, https://doi.org/10.15454/1.5572318050509348E12 City/Town/Region: Marcenat, Cantal Country: France Latitude and longitude (and GPS coordinates, if possible): 45° 19' 0.01" and 2° 49' 59.99"
Data accessibility	Repository name: Recherche Data Gouv https://doi.org/10.57745/H2SPNR
Related research article	None

Introduction

Drinking is an essential behaviour for the immediate survival of animals and therefore for their welfare (Welfare Quality, 2023). Drinking helps to compensate for the body water that gets used for digestion, waste excretion, thermoregulation, and growth. In dairy cows, water is especially important as it is used for milk production. Indeed, 34% of the water a cow ingests during the day goes into its milk. On average, one kilogram of milk is made up of 875 g of water, which equates to 26 L of water per day for a dairy cow producing 30 kg of milk per day (Holter and Urban, 1992). A dairy cow will drink from 15 to 120 L of water depending on environmental conditions (e.g. ambient temperature, access to water, etc.), physiological stage of the cow, DM content of the diet, social hierarchy within a group of cows, and other factors (Cardot et al., 2008; Coimbra et al., 2012; Khelil-Arfa et al., 2012; Lainez and Hsia, 2004). For instance, cattle increase their daily water intake as the ambient temperature or temperature humidity index increases (Beatty et al., 2006; Kadzere et al., 2002; McDonald

et al., 2020). In the context of climate change, environmental conditions leading to heat stress in dairy cows will become more and more common around the world due to global warming (North et al., 2023), making water accessibility to cattle a crucial issue on farms. Meeting the water requirements of dairy cows is therefore essential to ensure animal cow welfare and optimise milk performance especially in the context of climate change. Absence of thirst in dairy cows, one of the components of the animal welfare, is mainly deduced from environmental measures, such as number of functioning drinking troughs or cleanliness of the troughs (Welfare Quality, 2023). However, animal-based measures would be preferable to really capture the animal's point of view.

Visual observation remains the benchmark method for assessing drinking behaviour (Huzzey et al., 2005). However, although dairy cows housed in a free-stall barn spend little time daily on drinking (around 10 min per day, Nizzi et al., 2024), that drinking time is spread over the day in a number of drinking bouts (around 7 per day, Cardot et al., 2008). Day-long visual observation of the cows is therefore too time-consuming for routine practice. Intelligent recognition of basic behaviours (such as drinking) based on deep learning applied to video recordings is a good alternative to time-consuming visual observations (Wu et al., 2021). However, unlike methods based on on-board sensors, these methods have difficulty to identify which cow is drinking. Collar-attached accelerometers are already used on farms, where they show moderate to very good performances for detecting drinking (Tian et al., 2021; Williams et al., 2020) depending on the sensors and methods used. Electronic water bins connected to Radio-Frequency Identification ear-tags (Chapinal et al., 2007; Oliveira et al., 2018) and the combination of Ultra-Wideband indoor location with accelerometers (Benaissa et al., 2023) are currently the most effective tools for detecting drinking bouts in dairy cows.

Reticulorumen temperature (RT) sensors recorded temperature every 10 min have also been used and have shown promising results for detecting drinking bouts (Shirley et al., 2025; Vázquez-Diosdado et al., 2019). RT sensors detect drinking bouts based on the fact that RT falls when cows are drinking water (which most of the time will be colder than the ruminal content). The RT sensor-based drinking detection methods proposed by Vázquez-Diosdado et al. (2019) and Shirley et al. (2025) presented very different performances: less efficient than other sensors, affording only 74% of sensitivity (Se) and 73% of positive predictive value (PPV) (Vázquez-Diosdado et al., 2019), but more efficient for Shirley et al. (2025) (Se = 98%; PPV = 99%). It is still necessary to confirm the most suitable method for detecting drinking events based on RT data.

RT sensors are already used in climate change studies focused on heat stress in cows (e.g. Lees et al., 2019; Sharpe et al., 2021), as RT reflects core body temperature and is thus a good indicator of heat stress in cows (Ammer et al., 2016). Moreover, detecting drinking with a RT sensor involves the use of and data processing from only one sensor. In contrast, some other methods use combined technologies, such as Radio-Frequency Identification ear tags combined with electronic water bins (Chapinal et al., 2007) or accelerometers combined with Ultra-Wideband indoor location (Benaissa et al., 2023). RT sensors could therefore be a unique and useful tool for monitoring both heat stress and drinking behaviour if new methods can be developed to further improve their drinking detection performances. This would reinforce the role of the RT sensor in Precision Livestock Farming.

The aim of this study was to identify the most suitable method for detecting drinking bouts in dairy cows using sensors that recorded RT every 5 min. First, we tested three methods and compared their performances in terms of Se, PPV, and F-score. Two of these three methods were based on RT thresholds, i.e. 'FixT', based on a fixed RT threshold, and 'Cow-dT', based on a cow-day-specific

RT threshold (Vázquez-Diosdado et al., 2019). The third method, developed here, was based on RT fall slope ('FallST'). Second, we investigated detection performances according to drinking-bout characteristics (e.g. number of drinking events) for the best method. Third, we determined the accuracy of timing detection (how closely in time detected bouts matched to observed bouts). Finally, we investigated whether RT curve parameters (e.g. area under the curve) could predict drinking-bout characteristics (e.g. bout duration).

Material and methods

Animals, housing, and management

This experiment was carried out at the Herbipôle multidisciplinary experimental research platform (Herbipôle, INRAE), an upland ruminant farming systems research facility in Marcenat, France (<https://doi.org/10.15454/1.5572318050509348E12>). The study ran from 5 February to 21 March 2019. We used 28 primiparous Holstein dairy cows [age (mean $\pm$ SD) 3.2 ± 0.2 years] that were between 100 and 163 (128.9 ± 15.8) days in milk, with a body condition score of 2.5 to 3. The cows were loose-housed together in a cubicle barn (244 m^2 ; space allowance per cow: 8.71 m^2) counting 28 cubicles and 28 self-locking barriers, an automatic rotating brush (DeLaval Sweden) and a water trough (length: 222 cm). The water trough was located on the feed-access and exercise area (Fig. 1). It was equipped with a heating resistance unit to prevent water freezing and regulate the water trough temperature. Parasitic currents from the trough were neutralised by earthing. The water temperature was $10.7 \text{ }^\circ\text{C} \pm 2.0$ (mean $\pm$ SD), recorded 2 years after the study in the same conditions (same period, same trough, same barn with the same number of dairy cows housed). The building was lit from 0500 to 2000 h, and dimmed lights were left on at night. Cows were fed once per day (at 1000 h) with a total mixed ration based on hay (11.3 kg DM per cow), beet molasses (0.4 kg DM per cow), wrapped hay (5.8 kg DM per cow), concentrate (2.7 kg DM per cow, including nitrogen corrector and energy corrector), and minerals designed to meet the dietary require-

ments for lactation. The cows also had access to a salt lick. Any uneaten total mixed ration was pushed back towards the cows 3 times a day (at 1300, 1600, and 2200 h). The cows were milked twice a day (at around 0730 and 1630 h) by two experienced stockpersons in a 2×14 milking parlor adjacent to the barn.

Behavioural observations

To serve as a reference dataset to validate the three detection methods tested in this study, video-based observations of drinking behaviour were carried out using The Observer XT v14 behavioural coding software (Noldus, Wageningen, The Netherlands). The camera was purpose-positioned in the stall in a place overlooking the water trough. The clock of the cameras was synchronised ($\pm 2 \text{ s}$) to coordinated universal time standard. The 28 dairy cows were continuously video-recorded over two periods of 48 h (called 'series'). Two observers performed the observations, with each observer scoring one 48 h period, for a total of 96 h of observations (i.e. 4 days). The first observer was a technician in animal behavioural science with 22 years of experience in video-based behaviour scoring. The second observer was a student veterinarian trained in the task by the first observer. Inter-observer reliability on drinking behaviour duration per cow was excellent (Pearson correlation coefficient = 0.96).

Drinking behaviour was defined as "a cow fully or partially submerging her muzzle in water to consume water" (Genther and Beede, 2013). A drinking bout was composed of drinking behaviour events (i.e. uninterrupted drinking behaviour) separated by less than 4 min (Dado and Allen, 1993, 1994). A drinking bout started with the first sip of water and finished when the cow stopped drinking and at least 4 min elapsed without any further drinking behaviour (Fig. 2). Each drinking bout observed was characterised by five different variables: (i) *Drinking bout effective duration*, i.e. the total duration of effective drinking behaviour recorded during a single drinking bout, in seconds; (ii) *Drinking bout duration*, i.e. the time interval between the beginning and the end of the drinking bout, in seconds; (iii) *Interval to the previous drinking bout*, i.e. the time interval between the end of the previous drinking bout and the beginning of the current drinking bout, in minutes; (iv) *Interval to the next drinking bout*, i.e. the time interval between the end of the current drinking bout and the beginning of the next drinking bout, in minutes; (v) *Number of drinking events* (n/drinking bout), i.e. the number of separate drinking behaviours observed over the course of the whole drinking bout. Over the 96 h of observation, a total of 730 drinking bouts were observed (Table 1). Median effective drinking bout duration was 83 s (range: 1 to 781 s). Median drinking bout duration was 107 s (range: 1 to 1 289 s). Median interval to the previous drinking bout was 140 min (range: 4.1 to 1 056 min). Median number of drinking events was 3 (range: 1 to 33 drinking events per drinking bout).

Detection methods

Reticulorumen temperature sensor

Each cow was equipped with a reticulorumen sensor (Large Thermobolus, ITK/NewMedria, Châteaubourg, France), which was orally administrated to the cows using a bolus gun. The sensor recorded RT every 5 min (factory settings) with a precision of $0.1 \text{ }^\circ\text{C}$. The sensor gave an RT measurement every 5 min, along with a corrected temperature (raw RT processed to eliminate the effect of water drinking) (Fig. 3). The bolus data were transmitted to a local internet-connected gateway by LoRa (Long Range) technology and thence to FarmLife® web application (ITK/NewMedria, Châteaubourg, France), from where it was regularly downloaded for analysis and curation.

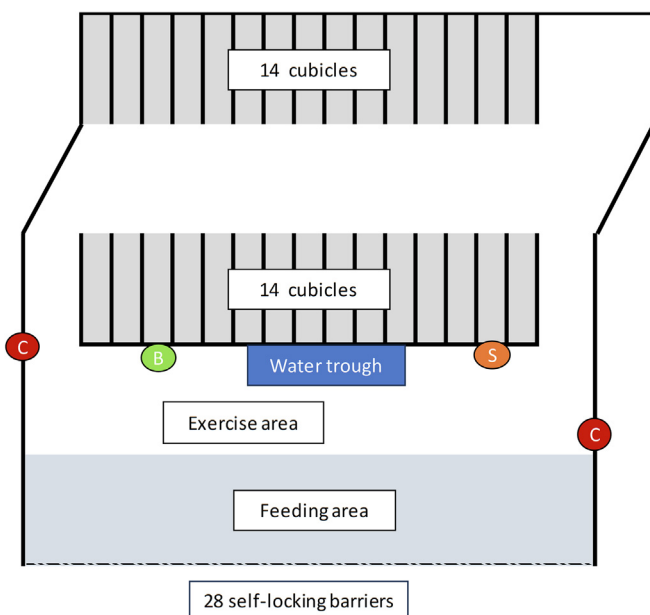


Fig. 1. Plan of the loose-housing cubicle barn and location of the water trough for 28 Holstein cows. C: cameras; B: brush (automatic); S: salt lick.

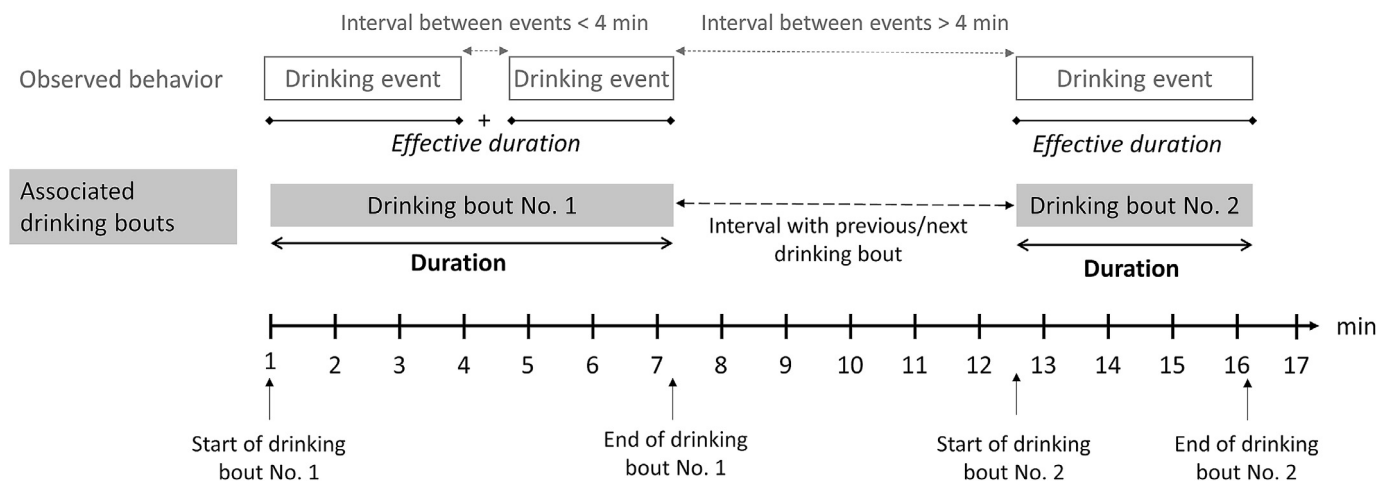


Fig. 2. Diagram of observed drinking behaviour and drinking-bout characteristics (effective duration, duration, interval between drinking bouts, and number of drinking events per drinking bout) of Holstein cows.

Table 1
Characteristics of the full set of drinking bouts observed in 28 Holstein cows continuously video-recorded over two periods of 48 h, for a total of 96 h of observations.

	n	First quartile	Median	Third quartile	Mean	SD	Min	Max
Drinking bout effective duration (s)	730	55	83	118	106	96.4	1.0	781
Drinking bout duration (s)	730	69	107	173	147	141.3	1.0	1 289
Interval from the previous drinking bout (min)	674	58	140	268	193	180.1	4.1	1 065
Interval to the next drinking bout (min)	674	58	140	268	193	180.1	4.1	1 065
Number of drinking events per drinking bout (n/drinking bout)	730	2.0	3.0	6.0	4.6	4.29	1.0	33

SD: standard deviation; s: second; min: minute; n: number of observed drinking bouts of the 28 cows over 96 h.

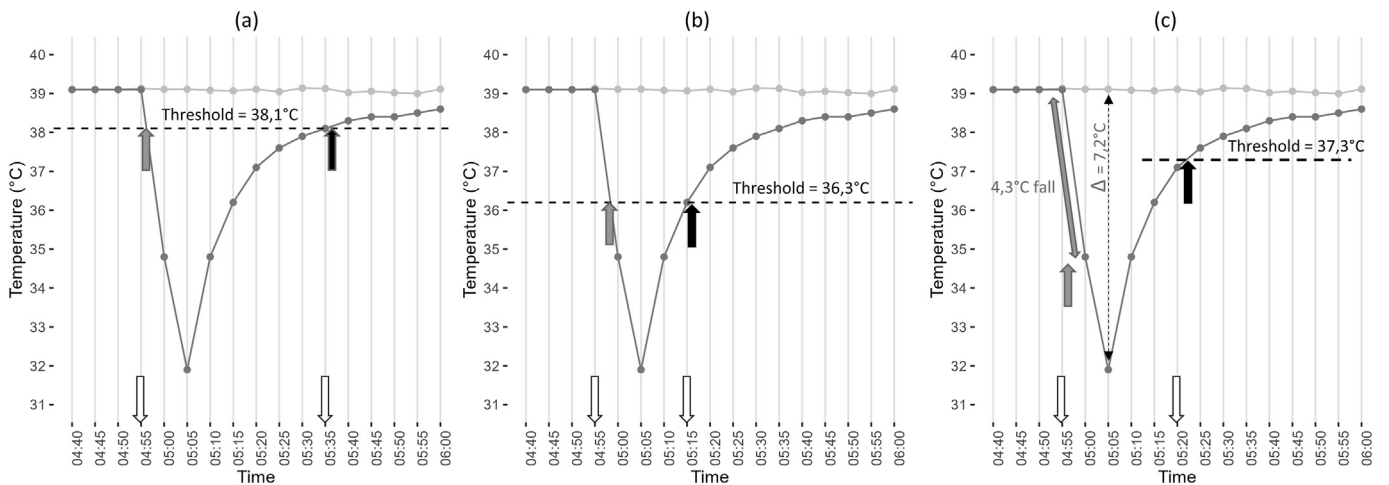


Fig. 3. Illustration of the detection of a drinking bout according to the three methods tested in this study: (a) FixT, based on a fixed threshold of 38.1 °C; (b) Cow-dT, based on a threshold calculated from cow-day data (in this example 36.3 °C); (c) FallST, based on reticulorumen temperature fall slope. On each graph, the vertical grey arrows indicate the change of reticulorumen temperature (RT) that led to drinking event detection (e.g. in (a) RT dropping below 38.1 °C). The black arrows indicate the moment that led to the detection of the end of the drinking bout (e.g. in (a) when RT rose up to 38.1 °C). The white arrows indicate the times corresponding to the beginning and the end of the detected drinking bout. On (c), Δ represents the RT fall due to drinking (the difference between the temperature just RT fell and the minimum RT reached during the drinking bout). Here, $\Delta = 7.2$ °C; thus, over the 0.5 °C temperature fall (within a 5-min interval) threshold, leading to the detection of a drinking bout. The threshold for detecting the end of the drinking bout with the FallST method is 37.3 °C (calculated as the sum of the minimum RT reached during the drinking bout, here 31.9 °C, and 75% of Δ , here 5.4 °C).

Fixed-temperature threshold method

The first detection method (FixT) was based on a fixed temperature threshold (adapted from Vázquez-Diosdado et al., 2019) by adding one criterion to detect the end of a drinking bout; (see below). The fixed threshold chosen was 38.1 °C, as this

threshold value provided the best detection performances in the study by Vázquez-Diosdado et al. (2019). A drinking bout was detected each time the RT fell below 38.1 °C. End of the drinking bout was determined as the time when RT returned to above 38.1 °C (Fig. 3).

Cow-day threshold method

The second method tested (Cow-dT) was the cow-day threshold method (adapted from Vázquez-Diosdado et al., 2019) by adding one criterion to detect the end of a drinking bout; (see below). A specific RT threshold for each cow for each day was calculated. The cow-day threshold corresponds to the mean of the RT measurements recorded for each cow over one day minus the standard deviation of RT of each cow on that day. The beginning of a drinking bout was detected when the RT fell below the cow-day threshold. The end of the drinking bout was detected when the RT returned above this threshold (Fig. 3).

Fall-slope threshold method

We also developed a new method, 'FallST', which is based on RT fall slope (similarly to Shirley et al., 2025, not published till after we completed our analysis). The beginning of a drinking bout was detected when the RT fell by more than 0.5 °C within a 5-min interval. The end of the drinking bout was determined as when the temperature fall recovered to 75% of the initial temperature fall due to drinking. These thresholds (0.5 °C and 75%) were defined empirically based on analysis of RT kinetics. For example, if the RT fell by 4 °C in a 5-min interval, then a drinking bout was detected as started, and when the temperature re-increased by 3 °C (i.e. 75% of the 4 °C fall), the bout was considered as ended (Fig. 3).

A criterion to detect the end of drinking bouts included in the three methods

To avoid missing a secondary drinking bout occurring before the RT returned to baseline, we developed a new criterion to detect the end of a drinking bout that was adapted for all three methods tested (Fix-T, Cow-dT, and FallST). In previous studies, the time required for RT to return to baseline varied from less than 20 min to up to 103 min depending on water temperature and quantity of water intake (Cantor et al., 2018). Moreover, RT can stay below the baseline temperature for hours after a drinking bout (Bewley et al., 2008). Therefore, using only the criteria described above for the three methods (Fix-T, Cow-dT and FallST) to detect the end of a drinking bout could have led to the end of a drinking bout being detected several hours after the actual end of that bout. A second drinking bout occurring during this period would have been missed as the end of the previous drinking bout had not yet been detected. We thus defined a new criterion to detect the end of drinking bouts that we applied and used with all three methods tested here (Fix-T, Cow-dT and FallST). For all methods, if one temperature point during a detected drinking bout was higher than the previous and the next temperature data point, then the drinking bout detected was considered as 'ended' and the start of a second drinking bout was considered as detected if the conditions for detecting a drinking bout were met per the method used (FixT, Cow-dT or FallST).

Output of the detection methods

The three drinking bout detection methods (FixT, Cow-dT and FallST) were applied on the RT data for the 28 dairy cows over the 4 days (i.e. 96 h) of video observations using an algorithm developed in visual basic for applications in Microsoft® Excel (2019). For each detection method, the output of the algorithm provided the date and hour of the beginning and end of each detected drinking bout, the lowest RT reached during the drinking bout, and the area under the curve of a detected drinking bout. The area under the curve of a detected drinking bout corresponds to the sum of the differences between corrected temperature provided by the sensor (raw RT processed to eliminate the effect of drinking colder water) and the RT of all temperature datapoints

included in the detected drinking bout (i.e. between the beginning and the end of the bout).

Data processing and analysis

Matching detected drinking bouts to observed drinking bouts

We matched the observed drinking bouts to detected drinking bouts by associating each detected drinking bout with the nearest observed drinking bout within a ± 30 -min window, to allow for sensor time drift. The median time lag between observed and detected drinking bouts was then calculated per cow and series (pairs of days). The times of detected drinking bouts were corrected based on this median (time-corrected detected drinking bouts) to avoid having non-matching bouts that may have been caused by sensor time drift. We then matched the observed drinking bouts with the time-corrected detected drinking bouts according to the two following criteria: i) Detected drinking bouts were only definitively matched with the observed drinking bouts if they were within a ± 10 -min window; ii) If two detected drinking bouts filled the criteria for matching to one observed drinking bout, the closest detected drinking bout was matched to the observed bout, and the second detected drinking bout remained unmatched.

Performance of the methods

Each detected drinking bout that was matched to an observed drinking bout was classed as a true positive (TP). Each detected drinking bout not matched to an observed drinking bout was classed a false positive (FP). Each observed drinking bout not matched with a detected drinking bout was classed a false negative (FN). Sensitivity (Se) was calculated as the total number of TPs divided by the total number of observed drinking bouts. Positive predictive value, or precision, was calculated as the number of TP divided by the sum of TP and FP. F-score was calculated using the following formula: $F - score = 2 \times \frac{Se \times PPV}{Se + PPV}$. Out of the three detection methods tested (Fix-T, Cow-dT and FallST), the one with the best F-score performance was kept for the following steps. Moreover, to investigate the influence of drinking bout characteristics on the detection performances of the best detection method, we calculated Se, number of TPs and number of FNs according to observed drinking bout characteristics (effective duration, duration, number of drinking events, time interval to previous/next observed drinking bout). Se was considered high if $\geq 80\%$, moderate if $< 80\%$ and $> 60\%$, and poor if $\leq 60\%$.

Accuracy of timing detection for the best method

The actual time lag between observed and detected drinking bouts (from the best method) was investigated using the time lag between observed and detected bouts for all TP before the timing correction (see in *Matching detected drinking bouts to observed drinking bouts* section). The time lag for each matched pair of detected and observed drinking bouts was calculated as the beginning time of the observed drinking bout minus the beginning time of the detected drinking bout.

Prediction of drinking bout characteristics from RT curve parameters

Based on the best method for detecting drinking bouts, we attempted to predict drinking bout characteristics (effective duration, duration, number of drinking events) using the RT curve parameters calculated from the sensor, i.e. the lowest RT reached during the drinking bout, area under the curve, and the calculated drinking bout duration. We tested (i) linear regression, (ii) negative binomial regression, (iii) log-transformed linear regression, (iv) decision tree, and (v) decision tree with log transformation of the drinking bout characteristics. We used linear regression as it is one of the simplest models, and decision tree as it is one of the

most common machine learning tools. We added log transformation and negative binomial distribution based on the observed distribution of the drinking bout characteristics. For each of these prediction models, we performed hyperparameter tuning to select the best combination of predictors (allowing for first-order interactions) for regressions (i) to (iii), and we used the minimum number of observations required in a node to attempt a split and the maximum depth of the tree for the decision tree models (iv) and (v). We used a nested cross-validation approach. We used 20% of the dataset for the final evaluation, and each inner cross-validation split used 80% of the subset for training and 20% for hyperparameter tuning. Performance of the algorithms was evaluated using three metrics: i) relative root mean-squared error (**RRMSE**, in percentage), ii) root mean-squared relative error (**RMSRE**, in percentage), iii) modelling efficiency (**EF**, unitless): These metrics were calculated as:

$$\text{RRMSE} = \frac{\sqrt{\frac{\sum_{i=1}^n (P_i - O_i)^2}{n}}}{\bar{O}} \times 100$$

$$\text{RMSRE} = \sqrt{\frac{\sum_{i=1}^n \left(\frac{P_i - O_i}{O_i}\right)^2}{n}} \times 100$$

$$\text{EF} = 1 - \frac{\sum_{i=1}^n (P_i - O_i)^2}{\sum_{i=1}^n (\bar{O} - O_i)^2}$$

where P_i is the i^{th} prediction by the prediction model, O_i is the i^{th} observation, and $\bar{O}$ is the mean of the n observations. RRMSE is used to evaluate residuals. It has the advantage of making all contributions positive and penalising greater errors more heavily. RMSRE also addresses residual errors but uses relative errors to adjust the weight of errors relative to the observed value. For RRMSE and RMSRE, the nearer to zero they are, the better the prediction is. EF was chosen as a correlation measure of model efficiency. EF statistic values lower than 0 result from a worse fit than the average of the observations.

Calculations were performed in R 4.4 (R Core Team, 2024), using the 'rpart' package (Therneau and Atkinson, 1999) for regression trees calculations and the 'glmulti' package (Calcagno, 2020) to select model regressions.

Results

Performances of the detection methods

Out of the total 730 drinking bouts observed, the FixT method based on a fixed threshold of 38.1 °C detected 657 bouts (Table 2). The FixT method also detected 26 drinking bouts that were not observed (FP) and failed to detect 73 drinking bouts that were observed (FN). The FixT method had a Se of 90.0%, a PPV of 96.2%, and an F-score of 93.0%. The Cow-dT method based on a threshold calculated from cow-day data detected 646 bouts out of the 730 drinking bouts observed. The Cow-dT method also detected 10 drinking bouts that were not observed (FP) and failed to detect 84 drinking bouts that were observed (FN). The Cow-dT method had a Se of 88.5%, a PPV of 98.5%, and an F-score of 93.2%. The FallST method based on RT fall slope detected 659 bouts out of the 730 drinking bouts observed. The FallST method also detected 5 drinking bouts that were not observed (FP), and failed to detect 71 drinking bouts that were observed (FN). The FallST method had an Se of 90.3%, a PPV of 99.2%, and an F-score of 94.5%. FallST thus showed the best F-score and was therefore selected for the following steps.

Detection performances according to characteristics of the observed drinking bouts (for the best method)

The FallST method performed poorly for detecting drinking bouts with an effective duration of ≤ 30 s (Se = 33.8%) but performed very well for detecting drinking bouts with an effective duration of > 30 s (Se = 96.4%) (Table 3). More precisely, Se was 17.1% for drinking bouts of [0–10] s, 30.8% for [10–20] s, and 60.9% for [20–30] s (Table S1, in supplementary material). Se reached 88.9% for effective durations of [30–40] s. The FallST method performed poorly for detecting drinking bouts with a duration of ≤ 40 s (Se = 42.3%) but performed very well for detecting drinking bouts with a duration of > 40 s (Se = 95.4%) (Table 3). Se got better as duration got longer: Se was 19.4% for drinking bouts of [0–10] s, 28.6% for [10–20] s, 50.0% for [20–30] s, and 73.9% for [30–40] s. The FallST method performed moderately well for detecting drinking bouts preceded by another drinking bout by an interval of 30 min or less (Se = 76.3%) but performed very well for detecting drinking bouts preceded by another drinking bout more than 30 min earlier (Se = 92.5%) (Table 3). The FallST method performed poorly for detecting drinking bouts followed by another drinking bout at an interval of 30 min or less (Se = 54.6%) but performed very well for detecting drinking bouts followed by another drinking bout more than 30 min later (Se = 95.8%). The FallST method performed moderately for detecting drinking bouts composed of a single drinking event (Se = 78.2%) but performed very well for detecting drinking bouts composed of two or more drinking events (Se = 92.4%).

Time accuracy of detection (best method)

The time interval between the beginning of observed drinking bouts and bouts detected by FallST ranged from –6.9 min to 10.7 min, with an average of 2.6 min. The confidence interval for the whole sample (mean $\pm$ 1.96*standard deviations) was [–2.9 – 8.1 min].

Performances of models for predicting drinking bout characteristics from RT curve parameters

All regression models showed very poor performance (Table S2, in supplementary material). The average RRMSE of the models tested ranged from 68.2 ± 5.1 (mean $\pm$ SD) to 110.8 ± 7.2 . The average RMSRE of the models tested ranged from 99.8 ± 66.6 to 394.6 ± 349.8 . EF ranged from -0.63 ± 0.13 to 0.32 ± 0.08 . The best model for predicting effective drinking bout duration was the decision tree with log transformation of drinking bouts, as it had the lowest RRMSE (68.2 ± 5.1) and EF (0.28 ± 0.16). Among the models predicting duration of drinking bouts, the decision tree regression had the lowest RRMSE (75.9 ± 7.2) and EF (0.32 ± 0.08) but with a very high RMSRE (394.6 ± 349.8). Note that the decision tree with log transformation of drinking bouts had similar results but with a better RMSRE (RRMSE: 77.7 ± 6.8 , RMSRE: 198.8 ± 135.7 , EF: 0.28 ± 0.11). The best model for predicting the number of drinking events was the decision tree with log transformation of drinking bouts, as it had the lowest RRMSE (80.0 ± 6.2) and EF (0.11 ± 0.08).

Author's point of views

The method developed here based on the fall slope of reticulorumen temperature (FallST, Se = 90.3%, PPV = 99.2, F-score = 94.5%) outperformed the fixed-threshold method (FixT, Se = 90.0%, PPV = 96.2%, F-score = 93.0%) and the cow-day threshold method (Cow-dT, Se = 88.5%, PPV = 98.5%, F-score = 93.2%) for detecting drinking bouts. This FallST method can accurately detect dairy

Table 2
Performances of the three drinking detection methods tested (FixT, based on a fixed threshold of 38.1 °C; Cow-dT, based on a threshold calculated from cow-day data; FallST, based on reticulorumen temperature fall slope).

Methods	Number of drinking bouts observed	TP	FP	FN	Sensitivity (%)	Precision (%)	F-score (%)
FixT	730	657	26	73	90.0	96.2	93.0
Cow-dT	730	646	10	84	88.5	98.5	93.2
FallST	730	659	5	71	90.3	99.2	94.5

TP: true positive; FP: False positive; FN: False negative.

Table 3
Performance of the drinking detection method based on reticulorumen temperature fall slope (FallST) according to observed drinking bout characteristics of 28 Holstein cows on 96 h period.

	N	TP	FN	Sensitivity (%)
Drinking bout effective duration				
≤30 s	71	24	47	33.8
>30 s	659	635	24	96.4
Drinking bout duration				
≤40 s	71	30	41	42.3
>40 s	659	629	30	95.4
Interval to previous drinking bout				
0–30 min	97	74	23	76.3
>30 min	577	534	43	92.5
Interval to next drinking bout				
0–30 min	97	53	44	54.6
>30 min	577	553	24	95.8
Number of drinking events				
1	110	86	24	78.2
>1	620	573	47	92.4

n: number of observed drinking bouts; TP: True positive; FN: False negative.

cow drinking bouts lasting more than 30 s and at least 30 min apart, with a detection time accuracy of 10 min. However, the models using RT curve parameters failed to predict characteristics of the drinking bouts (effective duration, duration, and number of drinking events).

The performances (Se, PPV and F-score) of three methods based on RT for detecting drinking bouts were better than another method based on collar-attached accelerometers (Williams et al., 2020) but slightly lower than a method based on electronic water bins connected to Radio-Frequency IDentification ear-tags (Chapinal et al., 2007; Oliveira et al., 2018) and a method combining accelerometers and geomagnetic sensors (Tian et al., 2021) or accelerometers and Ultra-Wideband location (Benaissa et al., 2023; Tian et al., 2021). Our methods had better F-Scores than in Williams et al. (2020) who used collar-attached accelerometers (F-score of 77% for the detection of drinking bouts lasting more than 10 s). The Se and PPV of our three methods (FixT, Cow-dT and FallST) were almost as good as in Tian et al. (2021) (Se = 94% and PPV = 98%) who used accelerometers combined with three-axis geomagnetic sensors. The best all-round performance for detecting drinking bouts is achieved by electronic water bins connected to Radio-Frequency IDentification ear-tags, as shown in Chapinal et al. (2007) (Se = 99.76%) and in Oliveira et al. (2018) (Se = 98.7%, PPV = 98.6%, and F-score = 98.7%). Detection using connected electronic water bins appeared to be slightly more effective than our RT-based approach; nevertheless, our method achieves satisfactory performances. Moreover, Chapinal et al. (2007) and Oliveira et al. (2018) methods require a complex electronic equipment that restricts access to water to one cow at a time per water point. Benaissa et al. (2023) and Tian et al. (2021) methods require two pieces of equipment to combine location and accelerometer data. Our system is simpler because it requires just one sensor (RT sensor) and works with all types of inside or outside drinking

troughs (individual drinking troughs, collective water troughs, natural water points, ...). Using RT sensor, it is possible to monitor cows drinking in any watering condition.

The three RT-based methods tested here performed better than the RT sensor-based methods used in Vázquez-Diosdado et al. (2019) but slightly lower than the method used in Shirley et al. (2025). Vázquez-Diosdado et al. (2019) tested two detection methods: (1) one with a general fixed threshold, and (2) one with a cow-day-specific threshold. Whatever the method used, their drinking bout detection performances were never better than 74% Se and 73% PPV, whereas here, we achieved Se and PPV values close to or better than 90% with both FixT and Cow-dT. Thus, even if FixT and Cow-dT were inspired from Vázquez-Diosdado et al. (2019), we found better detection performances. This apparent improvement is likely due to the different matching process used here compared to Vázquez-Diosdado et al. (2019). Vázquez-Diosdado et al. (2019) labelled each RT datapoint below the temperature threshold as 'drinking' and each RT datapoint over the threshold as 'not drinking', and then matched these points to behavioural observations, which means that each RT datapoint could be either TP, TN, FP, or FN. Here, we considered the detected drinking bout *per se* and not the individual sensor datapoint (each detected drinking bout was matched or not with a full observed drinking bout regardless of the number of RT datapoint it contained), which means that each detected drinking bout could be either TP or FP and each observed drinking bout could be either TP or FN. We think that the matching process used in Vázquez-Diosdado et al. (2019) was more likely to produce FPs, because RT variations are not strictly synchronised with observed drinking behaviour (e.g. RT takes longer to return to baseline than in a real drinking bout, thus creating FP points). Shirley et al. (2025) detected drinking events by using a method based on the same principle as ours (RT fall slope >0.5 °C). However, they achieved better performances

(Se, PPV, F-Score >98%). The difference in how 'drinking bouts' are defined could have contributed to the performance differences between our study and Shirley's study. Based on [Dado and Allen \(1993\)](#), we considered that a second drinking event was distinguished from the first if it occurred at least 4 min later vs 30 min in [Shirley et al. \(2025\)](#). We can suppose that the longer interval used in [Shirley et al. \(2025\)](#) has sometimes led to only one observed drinking bout, whereas in our study, we would have considered several drinking bouts. Therefore, the longer interval between drinking bouts in [Shirley et al. \(2025\)](#) would have contributed to fewer missed drinking events (FN) (especially those close together) and thus, a better Se. Nevertheless, our study confirms the efficiency of this method in detecting drinking bouts, on a larger sample size (730 vs 177 drinking bouts) and under different conditions to those of [Shirley et al. \(2025\)](#).

Our FallST method based on RT fall slope performed slightly better than the FixT method based on a fixed threshold and the Cow-dT method based on cow-day-specific threshold. To detect drinking bouts in cows, we compared a method based on a RT fall slope threshold (FallST) using RT sensors against the two methods adapted from [Vázquez-Diosdado et al. \(2019\)](#) based either on a general fixed threshold (FixT) or a cow-day specific threshold (Cow-dT). All three methods performed well (F-score > 90%), but FallST had a slightly better F-score (94.5% than FixT (93.0%) and Cow-dT (93.2%). FallST also presented fewer false negatives (FN = 84 for Cow-dT and 73 for FixT vs 71 for FallST) and fewer false positives (FP = 26 for FixT and 10 for Cow-dT vs 5 for FallST). Both FixT and Cow-dT detected a drinking bout each time the RT fell below 38.1 °C or below the cow-day threshold, and so they may have falsely identified drinking bouts after a very small drop in RT following the ingestion of any feed colder than the body). Indeed, RT was found to decrease after a cold meal (feed temperature from 1 to 20 °C), and this decrease was influenced by meal size ([Serviento et al., 2024](#)). The method developed here (FallST), based on slope of the fall in RT, marks an innovation compared with threshold-based methods (FixT and Cow-dT). FallST should improve detection, whatever the cows' farm system (pasture-based farming, feeding practices) or clinical health. For instance, in cows experiencing hyperthermia, we can reasonably assume that the FixT and Cow-dT methods might miss a drinking bout whereas FallST would probably detect it, as it is based on the absolute fall in RT.

The FallST method achieved high sensitivity (>92%) for drinking bouts lasting more than 30 s (90% of observed bouts) and at least 30 min apart from another drinking bout (86% of observed bouts). Se increased with effective duration of observed drinking bouts from less than 10 s (Se = 17.1%) to up to 20–30 s (Se = 60.9%) afterwards, and its Se was always >80% for drinking bouts lasting more than 30 s. We can reasonably assume that the drinking bouts lasting less than 30 s not detected in our study represent a very low percentage of the cows' daily water intake and therefore have little biological relevance. Nevertheless, in certain circumstances, such as heat stress or high competition at water points, cows can have close consecutive drinking bouts; thus, the absolute frequency of detected bouts should be treated with caution under these conditions. Water intakes in those short drinking bouts may be not enough to drive a decrease in RT. Another explanation could be related to the sampling interval between temperature data points, which was 5 min in the present study. Indeed, it is possible that this interval was too long to detect small temperature changes due to a small amount of water ingested. Previous studies have shown that the quantity and temperature of water ingested influence the kinetics of RT decrease, with water intake being positively correlated to the drop in temperature ([Ammer et al., 2016](#); [Cantor et al., 2018](#)). For instance, [Ammer et al. \(2016\)](#) showed that the mean fall in RT was 1.8 °C for water intakes <5 L but 4 °C for water

intakes >15 L (unknown water temperature). However, it is still unknown whether very low water intakes (e.g. <1 L) can still induce a fall in RT, including whether the water is warm or hot. Here, then, it is thus possible that water intakes in the drinking bouts that lasted less than 30 s were too low to induce a decrease of more than 0.5 °C in RT. We ruled out water trough temperature as a factor, as it was close to 10 °C which can be considered cold. The minimum water intake at a given temperature that is detectable based on RT remains to be investigated.

The FallST method showed very good sensitivity (Se > 92%) for detecting drinking bouts that were more than 30 min apart from another drinking bout. Se increased as the interval between the next or previous drinking bout increased, up until reaching 30 min. At intervals longer than 30 min, Se remains >85%. The time taken for RT to return to baseline (i.e. the temperature before the drinking bout) can range from a few minutes to hours ([Ammer et al., 2016](#); [Bewley et al., 2008](#); [Cantor et al., 2018](#)). It is highly possible that a second drinking bout occurring while RT is still low does not create a secondary fall in RT and thus cannot be detected using our method. Here, however, such drinking bouts represented a small percentage (14%) of the drinking bouts observed, and thus caused few FN.

The FallST method detected drinking bouts to within 10 min of the actual observed drinking bouts, which can be considered accurate. Thus, the timing of detected drinking bouts can be used to calculate the latency of drinking after a specific event. For example, it is known that cows drink a large proportion of their daily water intake after milking ([Cardot et al., 2008](#)), and so the time lag (allowing for the time precision of 10 min) to the first drinking bout after milking could be a good indicator of water accessibility.

Five types of models (linear regression, negative binomial regression, log-transformed linear regression, decision tree, decision tree with log transformation) were tested in an attempt to predict drinking bout characteristics from RT curve parameters, but they all gave poor results. The decision tree with log transformation gave slightly better results but was still unable to efficiently predict drinking bout characteristics. Here, we focused on simple models and simple metrics, but the very poor results suggest that further investigation should focus on other RT curve parameters and probably more complex prediction models. However, it may simply not be possible to predict drinking bout characteristics from RT curve parameters. To sum up, we were able to accurately detect drinking bouts in dairy cows, but without clearly determining the duration and number of associated events, which would probably require the use of other types of sensor data (e.g. accelerometers).

One of the main limitations of our study is that the detection method has only been tested in one situation (primiparous lactating Holstein-breed cows in free-stalls during winter, fed with total mixed ration, and a water temperature close to 10 °C). Further work should aim to test the FallST method in various situations that can influence rumen temperature (i.e. cows grazing on fresh/frozen grass) or the way RT responds to a drinking bout (i.e. various water temperature) in order to determine whether the method performs efficiently under all farming conditions. For the moment, we advise a validation effort by performing a few behavioural observations before applying the method to different experimental conditions. Moreover, attention must be paid in the case of using other devices especially if the sample frequency is different from our study. Indeed, our method is validated with a 5-min sampling frequency but it should be validated and adapted if the sensor has a different sampling frequency -e.g. 10-min sampling with the smaXtec bolus (smaXtec animal care GmbH, Graz, Austria).

Furthermore, our reference standard (behavioural observation through video recording) is probably not 100% reliable for

detecting actual drinking bouts. For instance, a low Se for short drinking bouts may also be due to our observation method. Our definition of a drinking bout was when “a cow is fully or partially submerging her muzzle in water to consume water” (from [Genther and Beede, 2013](#)). However, there may have been cases where a cow submerged her muzzle but did not really consume water, especially when the muzzle was submerged for only a short time (e.g. 5 s). To be sure that the cows really consumed water, behavioural observation could be combined with individual connected troughs.

Our method allows the detection of bout timing and frequency; nevertheless, we did not test if it can estimate drinking intake in terms of volume as our experimental design did not allow that (collective drinking trough). The next step should investigate if some characteristics of the temperature signals (e.g. are under the curve) of a drinking bout (considering water temperature) could be used to estimate water intake.

In conclusion, using the detection method developed here based on RT fall slope, we were able to efficiently detect drinking bouts in dairy cows. This method provided the frequency and accurate timing of drinking bouts using a unique sensor. However, we were unable to further characterise drinking bouts, such as drinking bout duration, based on RT curve characteristics. The RT fall slope method could probably be applied to all classes of cattle and other ruminants.

Peer Review Summary and Supplementary material

Peer Review Summary and Supplementary material for this article (<https://doi.org/10.1016/j.anopes.2025.100107>) can be found at the foot of the online page, in Appendix A.

Ethics approval

The protocol and procedures were approved by the local ‘CEMEA Auvergne’ institutional animal care and use committee (CE-05092.01). It was conducted in full compliance with the ethical standards of the institutional and national committee, all applicable provisions established by European Directive 2010/63/EU. The reference of the committee approval is APAFIS agreement #2015043014541577. All procedures were applied by trained staff members who performed the trial in accordance with all relevant guidelines and regulations. The study was carried out in accordance with the ARRIVE guidelines ([Percie du Sert et al., 2020](#)). All animals used in this study were handled in strict adherence to good clinical practices. Experimental cows went through a clinical examination performed by an experienced veterinarian. The clinical examination included checks for signs of lameness, nasal discharge, and respiratory disorder, and measurement of rectal body temperature, heart rate, respiratory rate, and rumen contraction rate. Endpoints and exclusion criteria were defined before the start of the experiment, as follows: any cows showing any signs of sickness or distress during the experiment were examined by a veterinarian and were removed from the study if they exceeded the threshold limit of rectal temperature above 42.5 °C for 4 consecutive hours, together with a score above 12 on the [de Boyer des Roches et al. \(2017\)](#) grid.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) did not use any AI and AI-assisted technologies.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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